

## Lab 04

### Hint:

1. **Correction:** "\*" and "\" are for matrix operation, ".\*" and ".\" are for element-wise operation .
2. For question about MATLAB syntax, search from MATLAB documentation first.
3. Read problem statement carefully, **word-by-word**. Check all the requirements.
4. Download **all files for the lab to one single folder** and make sure you are working on the folder with all the files, otherwise, you will get errors.
5. You **WON'T be able to open "\*.p" file (as always)**. Instead, read the problem statement for an example of using it. NO help file this time.
6. For linear system  $\mathbf{A} \cdot \mathbf{x} = \mathbf{b}$ ,  $\mathbf{x} \neq \mathbf{b}/\mathbf{A}$ .
7. Problem 2: directly edit imageRecon5x5.m downloaded from Canvas, no need to create a new one.
8. **Don't hard-code**
  1. **matrix P**, use nested for-loops.
  2. **dimension of P**, get it from **x**.
9. Make sure you **correctly identify the indices** for columns and rows
10. Problem 2.2: don't use "colormap('gray')" for plotting.

### Submission checklist (grading rubric):

- Problem 1.1: **tpnPlanner.m**
  - Complete header (system formulation described, recovered volumes stated) and commented code
  - Correct formulation and construction of **A** and **b**
  - Correct use of MATLAB's backslash operator (\) and delivered-totals check
- Problem 1.2: **tpnPlannerSub.m**
  - Correct substitute matrix
  - Header explanation of why the prescription cannot be met
- Problem 2.1: **imageRecon5x5.m**
  - Fully commented code and header
  - Correct construction of the matrix A
  - Correct use of MATLAB's backslash operator (\) to solve linear system
- Problem 2.2
  - Correct indexing into P
  - Correct **image figure** w/ title

### Expected results:

#### Problem 1.1:

x =  
1.0000  
0.5000  
0.5000

A\*x =

125  
50  
55

#### Problem 1.2:

x =  
NaN  
-Inf  
Inf

A\*x =

NaN  
NaN  
NaN

#### Problem 2.2:

